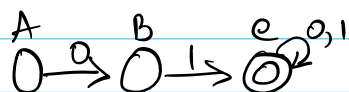
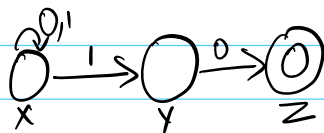


⑧ {String s | s start at 01 & end at 10}

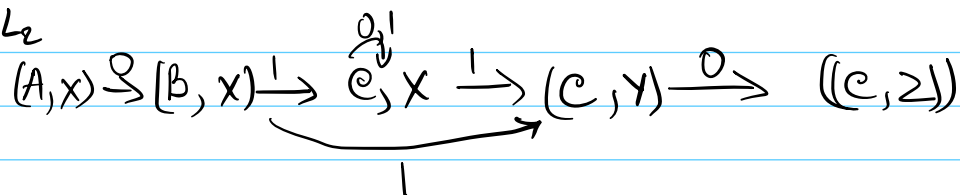
L_1 s start at 01



L_2 s end at 10



$L_1 \cap L_2$



Test begin A, X

0 $\rightarrow (B, X)$

01 $\rightarrow (C, X)$ or (C, Y)

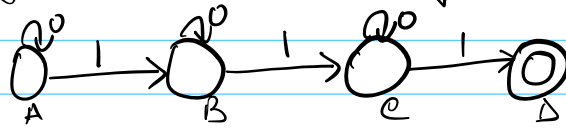
011 $\rightarrow (C, X)$ or (C, Y)

0110 $\rightarrow (C, X)$ or (C, Z)

01101 $\rightarrow (C, X)$ or (C, Y)

011010 $\rightarrow (C, X)$ or (C, Z)

(12) {String S | S contains exactly 3 1's}

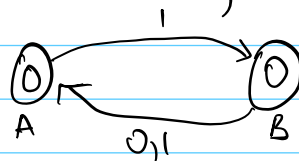


Begin A.

0 $\rightarrow$ A
 01 $\rightarrow$ B
 010 $\rightarrow$ B
 0101 $\rightarrow$ C
 01010 $\rightarrow$ C
 010101 $\rightarrow$ D
 0101010 $\rightarrow$ D
 01010101 $\rightarrow$ A

010101010 $\rightarrow$ B
 0101010101 $\rightarrow$ C
 01010101010 $\rightarrow$ C
 010101010101 $\rightarrow$ 1

(16) {String S | every odd position of S is 1, starting at 1.}



start ((A)) it's empty string

1 ((B))

10 ((A))

101 ((B))

1010 ((A))

10101 ((B))

if 10100 $\rightarrow$ reject (false)